



MASK
with
VIBRACHROM

TECHNICAL DOSSIER

davines

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1. LINE FEATURES AND BENEFITS

Mask with Vibrachrom™ - Davines innovative colouring system introducing technology new to the world of hair colour that allows superior, uniform colour penetration in to the hair's structure, offers extra shine, high conditioning effect and outstanding duration of the colour.

Mask with Vibrachrom™ is an ammonia-formulated, permanent colour offering professional hair colourists complete versatility whether covering white hair, reflect changing, darkening or lightening up to three levels. Additional lightening of up to four levels, and further creative possibilities, become possible with the use of Lightening Booster 000 and the Extralift series offers up to five levels of lightening.

Mask with Vibrachrom™ comes with a dedicated range of Activators (5, 10, 20, 30 and 40 volume).

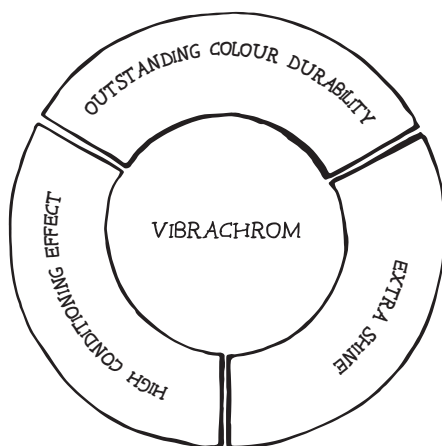


1.1 MAIN CLAIMS AND ACTIVE INGREDIENTS

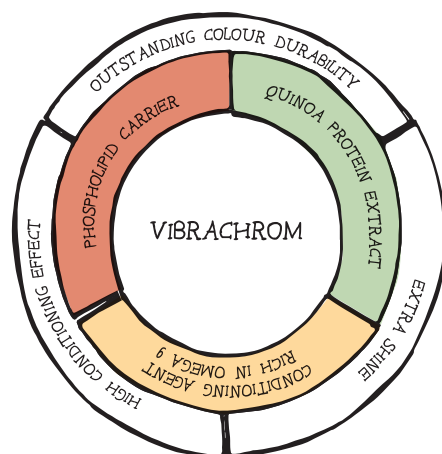
Thanks to innovative Vibrachrom™ technology, born from a crossover research of the natural, cosmetic and textile worlds, Davines Research and Development Laboratories can now offer the professional hair colourist and their clients.

EXTRA SHINE HIGH CONDITIONING EFFECT OUTSTANDING COLOUR DURABILITY

The result is extremely shiny hair with long-lasting colour fidelity.

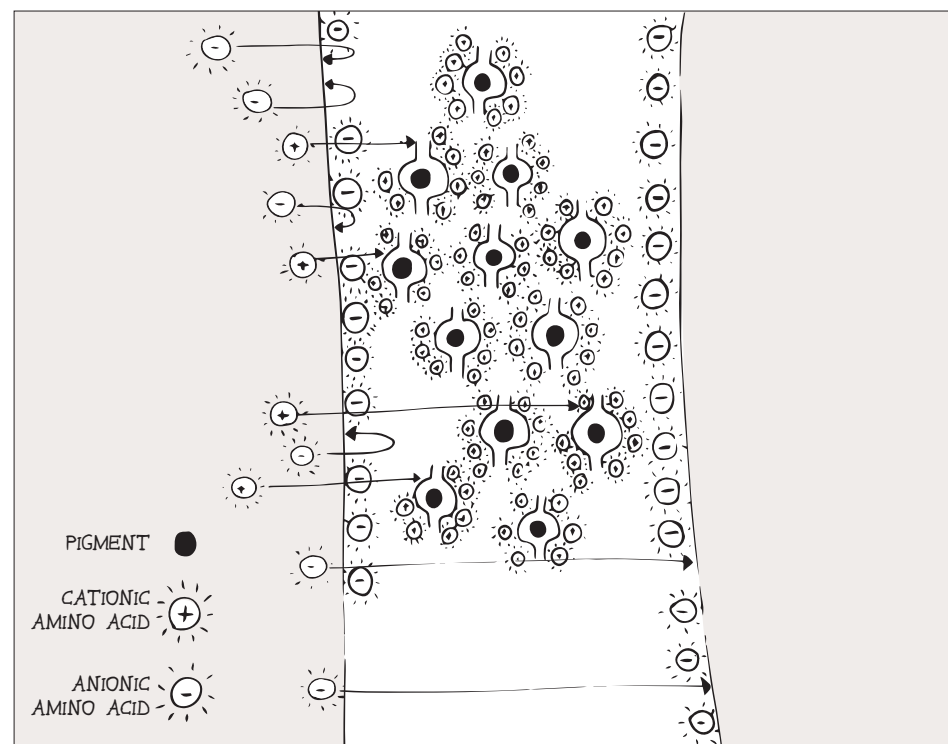


Vibrachrom technology incorporates:



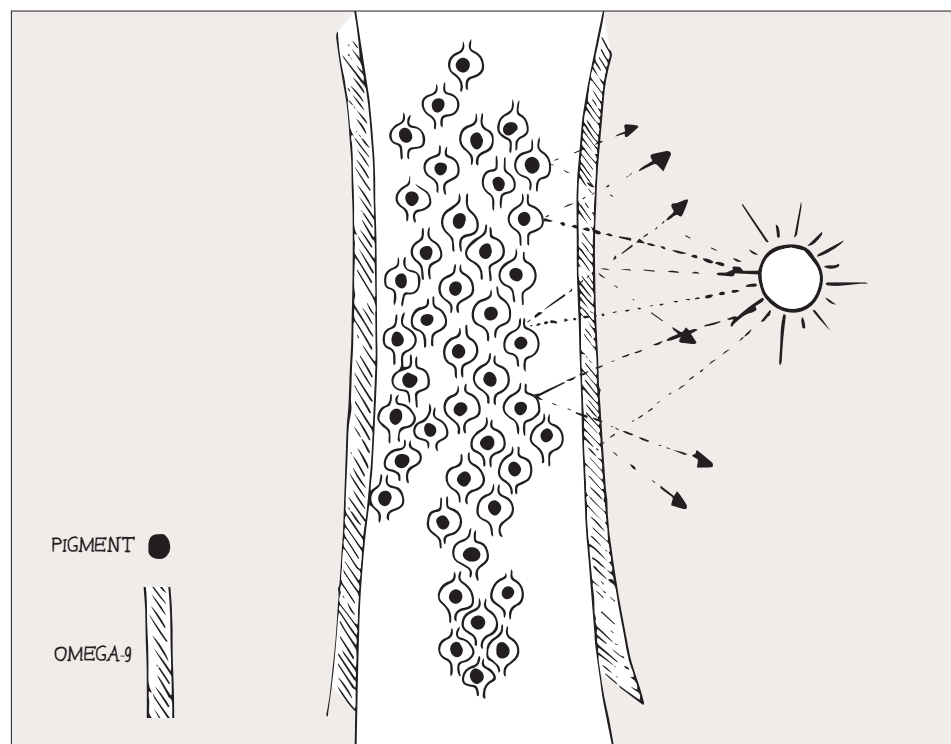
1. **A QUINOA PROTEIN EXTRACT**, with the power to increase colour absorption during the colour service and ensure that the hair fibre retains colour for longer. Quinoa is composed of cationic and anionic amino acids, providing dual-action benefits. Cationic amino acids are very similar to those found in the hair structure and anchor themselves to colour molecules, allowing improved uniting. Meanwhile, anionic amino acids coat and protect the hair resulting in longer-lasting colour. Anionic amino acids are also free radical scavengers that minimise oxidative damage and consequently increase colour fidelity and duration.

Quinoa (*Chenopodium quinoa chenopodiaceae*) has been cultivated since 3,000 B.C. and has always been a vital source of nourishment. This ingredient was considered sacred by the Incas, who called it the "Mother Grain". What makes this ingredient really important is the high protein content, containing all the eight essential amino acids and this makes its presence fundamental for our colour line. The quinoa that we selected to use in the production of Mask with Vibrachrom™ is grown in Peru respecting the local population and territory.



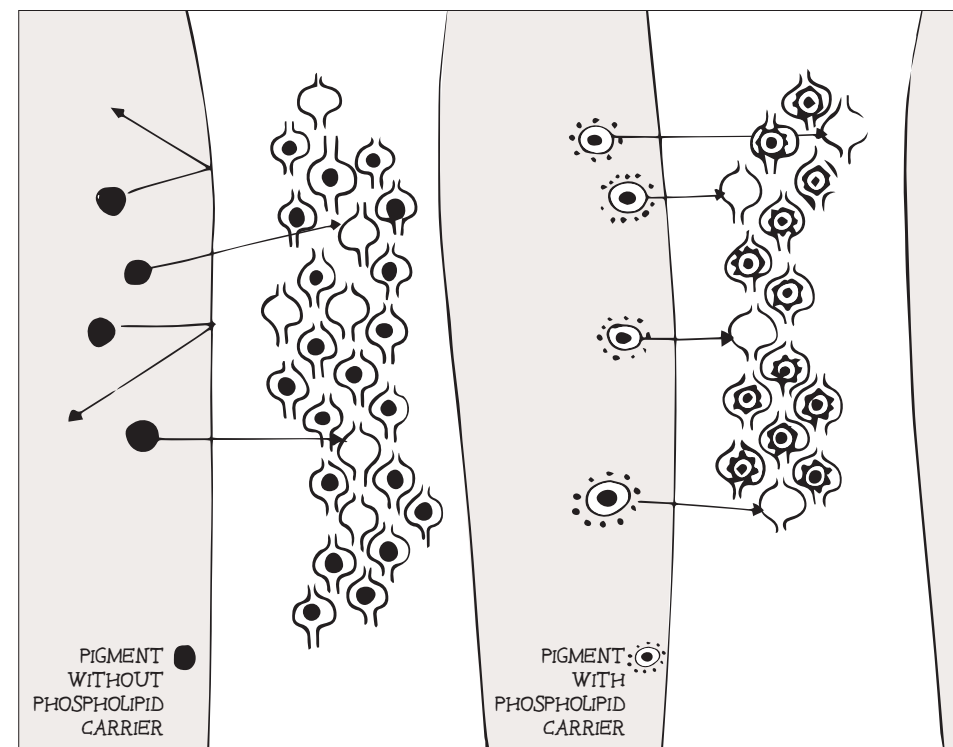
2. A CONDITIONING AGENT RICH IN OMEGA-9 (ω -9), vegetal derived, able to form a lipid film that protects, smooths and nourishes hair enhancing inner and outer light refraction: outer refraction, creating extra shine, is due to the compactness of the hair, while inner refraction relates to the transit of light through the pigment-containing cortex, similar to how light passes through a glass, and illuminates the hair colour, increasing vibrancy.

The potent efficacy of this conditioning agent is due to the high concentration of ω -9. Omega-9 is a natural derived fatty acid, extracted from plants particularly rich in it and, in addition to having nourishing and elasticizing properties to hair, also has a special branched structure, similar in form to a small "hair pin". This distinctive structure allows it to firmly attach itself around the hair thus strengthening the coating and protective effect.

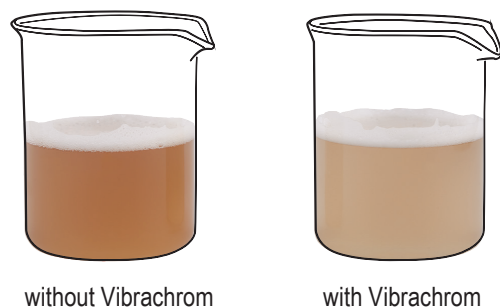


3. A PHOSPHOLIPID CARRIER created for the textile world and never used before in cosmetics, it increases pigment permeability by reducing surface tension. Thanks to its dispersant capacity, the phospholipid carrier guarantees a uniform migration of the colour molecules evenly throughout the hair length thus increasing shine, and equalizes colour more homogeneously even on particularly porous or resistant white hair. This ensures durable, longer-lasting colour.

In the process of research and innovation, Davines always tries to enlarge its spectrum of action and, for this reason, we naturally tend to explore what occurs in other industries where there is a relationship to our own. Considering the importance of colour in the world of textile production, we enquired and examined. We discovered that specific ingredients are used, phospholipid carriers, to increase the intensity of results even on delicate fibres such as silk by increasing the pigment's migration inside the fibres, making the colour appear more "full". We then went on to discover that the phospholipid carriers have the same efficacy also on the hair, enhancing pigment penetration.



Vibrachrom technology takes Mask colour system into new and improved levels of performance, particularly its capacity to reduce colour loss. This improved performance can be seen; it can also be demonstrated. Using standardized methodology, washing tests were conducted on hair strands using Mask formulas with and without Vibrachrom. 10 shampoos were conducted to measure colour retention by collecting all the rinse water. Comparing the water, the two samples in the following image clearly evidence the reduced colour loss with Vibrachrom.



Mask with Vibrachrom™ is formulated without PPD and parabens.

Mask with Vibrachrom™ works in synergy with complementary Davines **ACTIVATORS**, formulated in a cream emulsion enriched with conditioning agents that respect and protect the hair structure during colouration.

Emollient substances were also added to make the pigments more soluble, making it easier for them to penetrate the hair structure.

Activators come in five strengths: 5, 10, 20, 30 and 40 volume. Note that Mask with Vibrachrom™ must always be mixed with Davines Activators.

Activator	% H ₂ O ₂	EXPECTED RESULT
5 volume	1.5	Toning bleached or coloured hair, hair with highlights or refreshing natural hair, length and ends
10 volume	3	Darkening
20 volume	6	White hair coverage, darkening, reflect change, lightening up to 1 level
30 volume	9	Lightening up to 2 levels
40 volume	12	Lightening up to 3 levels and 5 levels with extra lift

1.2 LINE BENEFITS

1. FLEXIBLE, EASY-TO-USE SYSTEM

With a fully intermixable system of naturals and reflective nuances, hair colourists can customize their client's hair colour according to their desire by selecting the most suitable natural base (Natural, Natural Intense, Natural Ash, Natural Warm) employing one optimal mixing ratio and a single processing time.

2. IDEAL COVERAGE OF WHITE HAIR

With Vibrachrom™ technology comes improved pigment penetration and colour equalization. Four chromatic choices in the Mask Natural family, (Natural, Natural Intense, Natural Ash, Natural Warm) can be used alone or mixed with reflective nuances to satisfy all chromatic and coverage needs. The Intense Naturals, for example, should be selected in cases of difficult to cover, resistant white hair to guarantee perfect coverage and can particularly satisfy colour clients who desire more intense white hair coverage.

3. PERFECT LIGHTENING CAPACITY

All reflect nuances are chromatically formulated to contain only very limited quantities of base pigments thus allowing the reflective pigments to be more vibrant, strengthened, and optimizing the line's lightening capacity.

4. LOW AMMONIA SMELL

The pleasing scent of Mask with Vibrachrom™ actively disguises the smell of ammonia, for a more pleasant colour experience.

5. SUSTAINABILITY

Emissions from Mask with Vibrachrom™ packaging production are compensated by participating in reforestation projects. The quantity of packaging material used has been reduced as much as possible, eliminating any unnecessary material and thus optimizing transportation, further reducing CO₂ emissions. The packaging is made of FSC recycled paper and is compostable.





2. DIAGNOSIS

Before applying Mask with Vibrachrom™ it is advised to carry out a diagnosis of the client's hair with some important assessments that will ease the colourist's task and guarantee full customer satisfaction. Consider all of the following points:

- **Desired colour:** thanks to the valuable support of the Mask with Vibrachrom™ colour chart which contains useful information about colour, select with the client the desired colour or combination of colours.
- **Natural level:** it is important for the colourist to accurately assess the natural level of the hair. Use the Natural family swatches in the Mask with Vibrachrom™ colour chart as a guide. The difference between the natural level of the hair and the level of the desired hair colour is fundamental in selecting the most appropriate Activator strength to be used.
- **Residual pigments:** when lightening hair, simultaneously the hair is coloured and the natural pigments (melanins) are lifted. The term residual pigments refer to the melanins that are still present in the hair after colouring. Their composition varies according to the hair type and level of lightening achieved (see fig. below). The residual pigments will influence the final chromatic result.
- **Cosmetic pigments:** are the pigments from previous cosmetic colours.
- **Percentage of white hair:** it is necessary to assess the presence of white hair, as well as its distribution or presence only in specific areas. Assessing the correct percentage determines the choice of Activator and the colour mix.
- **Hair condition:** damaged hair, for example, is porous and thus absorbs colour more easily, but fades equally easily.
- **Scalp:** always check for irritation or reddening. Do not proceed with the colour service if irritation or reddening is observed.
- **Porosity and structure:** it is important to distinguish fine, medium or thick hair, assessing also its porosity because - as previously mentioned - this can influence pigment absorption. Fine hair is the easiest to colour, whereas thicker hair can be resistant to colour, especially if white hair. In case of resistant hair, consider increasing the processing time and apply the colour generously and precisely.
- **Hair type:** hair types can vary regionally and according to the individual or ethnic group. Generally, finer hair has a weaker structure and therefore standard processing times apply, whereas thicker hair can be more difficult to colour and an extended processing time should be considered.



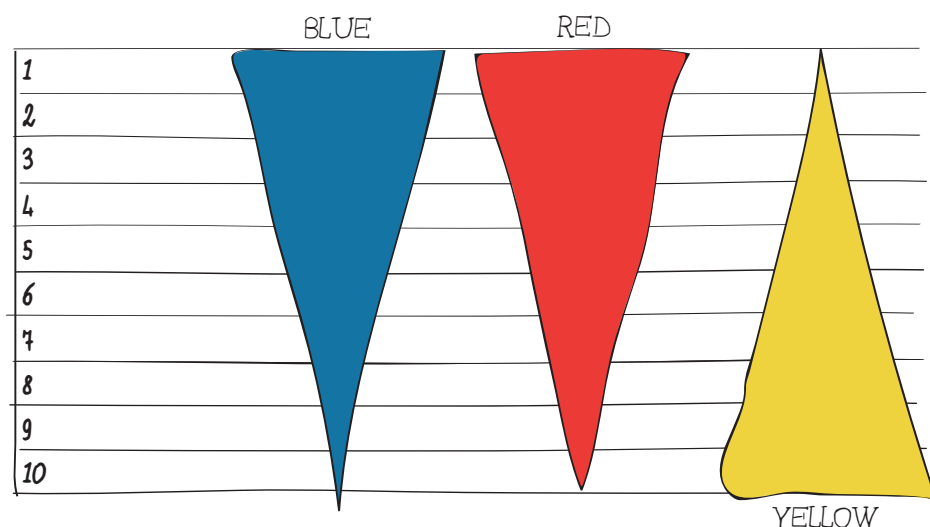
2.1 MELANIN

The natural colour of hair depends on the quality, quantity and distribution of the melanins it contains. The production and distribution of melanin changes with time and it depends on an individual's genetic makeup and physiological characteristics. It also changes with age and in white hair, melanin is totally absent.

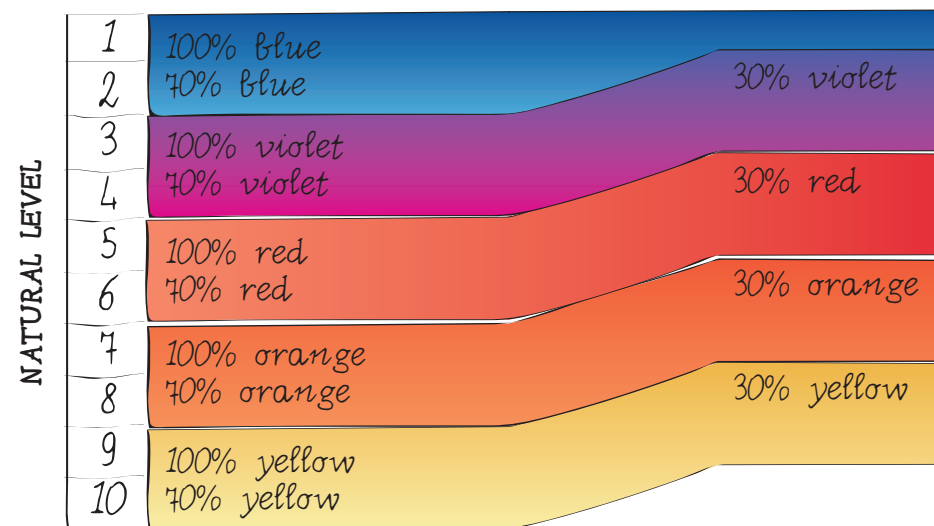
In nature, there are some types of melanins characterised by different reflects, and it is their varying proportion that defines the natural colour of hair.

When combined in equal quantities, primary colours theoretically give rise to black. In reality, what is obtained in nature is never black, but a shade of extremely dark grey that looks like black, because pigments, including melanins, lack a chemical structure allowing them to fully absorb light. Therefore, in theory, if hair had yellow, red and blue melanins in equal proportions, it would look grey. In nature, hair colours do not vary on shades of grey, but rather of brown because the various types of melanins are present in different proportions.

The chart below shows the likely distribution of melanins in the various levels. The term "level" refers to the intensity of the basic level, as we will see later, i.e. how light or dark is the hair's natural colour on a scale from 1 to 10. Blue and red tend to be predominant over yellow at lower levels, which is shown by the widest part of the triangles that corresponds to the level number, whereas yellow prevails in lighter levels.



The chart below shows instead the variation of the main reflect, i.e. the prevailing component amongst the types of melanins in the various levels, and its change following lightening, with the residual melanins highlighted.



For instance, if we analyze level 2, we will notice a prevalence of blue and violet. If we lighten level 2 hair by three levels, we will reach level 5, where the prevailing component is red, which will have to be taken into account since it affects the final result. For example, if you want to achieve a cold colour, you will need to use a nuance with a higher percentage of green to counteract the red resulting from lightening.

When carrying out a bleaching or an oxidation colour that leads to lightening, it is important to bear in mind that it is impossible to lift hair up to white because feomelanin, i.e. the yellow melanin, is always present.

In general, you should always take the contribution of residual melanins into account since it changes in terms of composition and amount according to the starting level and the lightening achieved.



3. SKIN SENSITIVITY TEST

It is always advised to carry out a skin sensitivity test 48 hours before applying the colour to avoid any allergic reaction. Follow the instructions on the leaflet accompanying every box of Mask with Vibrachrom™.

4. USING PROTECTION PROVIDER

Before applying the colour, it is advised to apply Protection Provider around the perimeter hairline, including the forehead and behind the ears. Protection Provider contains film-forming agents that protect the skin from possible staining. Delicately and precisely apply a thin layer of Protection Provider. Do not apply using gloves.

5. HOW TO USE

To prepare the mixture, weigh the colour and activator according to the results you wish to obtain. Mask with Vibrachrom™ should always be applied to dry, unwashed hair.

5.1 OPTIMIZING COVERAGE OF MORE THAN 50% WHITE HAIR

The Mask with Vibrachrom™ **Natural family (Natural - Natural Intense - Natural Ash - Natural Warm)** perfectly cover up to 100% white hair. The Mask with Vibrachrom™ Intense Cool Natural family perfectly covers 100% white hair. When Mask with Vibrachrom™ reflect nuances are used on more than 50% white hair, follow the instructions below, choosing the most appropriate Natural family and the correct activator to obtain the desired result.

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 White hair coverage: Natural, Natural Intense, Natural Ash and Natural Warm	 1:1,5  20	 50 g natural base + 75 g Activator 20 volume	 35 MIN	 20 MIN
 White hair coverage(>50%): Reflects	 1:1,5  20	 ***1/3 natural base + 2/3 reflect + 75 g Activator 20 volume	 35 MIN	 20 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary.

** It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal.

*** the mixing ration can be extended to 1/2 + 1/2 to increase coverage.

The flexibility of an open mixing ratio offers choice to satisfy all white hair coverage needs and varied hair types.

When mixing a Natural with a reflective nuance, choose the most suitable Natural family for the desired final result.

For example: a client has 100% white hair and wishes to obtain a 6,3. The Natural Warm 6 will support achieving the golden reflect of the 6,3. However, the professional hair colourist may choose to use 6,0 instead. The result of the reflect will be more towards the Natural and slightly less golden.

The **Natural Intense** family offers increased coverage: for fine/normal hair, we recommend using the Activator 20 volume with the standard mixing ratio, while for resistant hair, use Activator 20 volume but increase the natural base according to the table below.

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Covering particularly resistant white hair (>50%): Natural Intense and Intense Cool Natural	 1:1,5  20	 1/2 natural intense base (25 g) + 1/2 reflect (25 g) + 75 g Activator 20 volume	 35 MIN	 20 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary.

** It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal.

The Intense Cool Naturals family is specifically formulated to remain balanced on level and reflect for up to 6 weeks. They are enriched with natural oils to guarantee a rich coverage and colour result, with a smoother cream formula. Important tips when using intense cool naturals: When used on pigmented hair below 50% white, the result will appear darker due to the intense depth and extra cool reflect.

When used on non resistant white hair above 50%, the result may appear darker than expected. When used in foil the shade may appear darker than expected.

5.2 COVERAGE FOR LESS THAN 50% WHITE HAIR

All Mask with Vibrachrom™ reflect nuances perfectly cover up to 50% white hair without the addition of a natural base nuance.

Choose the desired nuance for the type of colouring to be achieved, following the mixing ratio:


 1:1,5  20

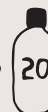
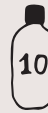
 35 MIN

**1 part colour +
1.5 parts activator
standard processing
time is 35 minutes**

For example: a client wishes to become a 6,3 and his/her natural level is a 6,0, if there is less than 50% white hair, the colourist will choose a 6,3 mixed with Activator 20 volume. Standard processing time of 35 minutes.

5.3 REFLECT CHANGING AND DARKENING

When reflect changing or darkening the hair, follow the instructions in the table below:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Changing reflect	 1:1,5  20	 50 g colour + 75 g Activator 20 volume	 35 MIN	 20 MIN
 Darkening	 1:1,5  10  1:1,5  20	 50 g colour + 75 g Activator 10 volume or Activator 20 volume	 35 MIN	 20 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary.

** It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal.

When darkening normal or fine hair, as a general guideline, Activator 10 volume can be selected, while Activator 20 volume is recommended for thick or particularly porous hair.

5.4 TONING

This colour service can be done both on previously-coloured hair to refresh reflects, to correct undesired colours, or to tone highlights as well as on virgin hair for first-time applications. Apply to damp, towel-dried hair and mix the desired nuance with Activator 5 volume according to the instructions below:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME*
 Toning	 1:2  5	 50 g colour + 100 g Activator 5 volume	 1-15 MIN

*standard processing time. Processing times can be lengthened if required, depending on the type of hair being treated.

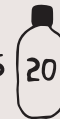
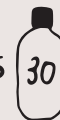
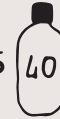
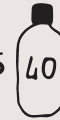
Processing times vary by application: for coloured hair or to tonalize highlights or previously bleached hair, we recommend a regular visual check. Instead, for virgin hair, processing times can extend up to 15 minutes.

5.5 LIGHTENING

When wishing to lighten natural hair it is important the correct Activator to mix with the desired colour to reach the desired lightening.

Remember that every Mask with Vibrachrom™ nuance can achieve up to three levels of lightening and, if mixed with lightening Booster 000, can obtain four levels.

Please follow the instructions in the table below:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Lightening up to 1 level	 1:1,5  20	 50 g colour + 75 g Activator 20 volume	 35 MIN	 20 MIN
 Lightening up to 2 levels	 1:1,5  30	 50 g colour + 75 g Activator 30 volume	 35 MIN	 20 MIN
 Lightening up to 3 levels	 1:1,5  40	 50 g colour + 75 g Activator 40 volume	 35 MIN	 20 MIN
 Lightening up to 4 levels	 1:1,5  40	***1/3 Lightening Booster 000 + 2/3 colour + 75 g Activator 40 volume	 35 MIN	 20 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary. **It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal. ***The mixing ration can be extended to 1/2 + 1/2 to increase the lightening ability.

5.6 EXTRALIFT SERIES

Mask with Vibrachrom™ Extralift series is formulated for extra lightening, up to 5 levels. The Extralift series can be also used with Activator 30 volume to achieve up to 4 levels of lightening. To obtain the desired result, it is recommended to use the Extralift series on level 6 natural bases and higher.

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Lightening up to 4 levels	 1:2  30	 50 g extralift + 100 g Activator 30 volume	 50 MIN	 30 MIN
 Lightening up to 5 levels	 1:2  40	 50 g extralift + 100 g Activator 40 volume	 50 MIN	 30 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary. **It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal.

5.7 LIGHTENING BOOSTER 000

This is a "Speciality", as its formulation is identical to all the others nuances but it does not contain pigments and is used to increase lightening.

Follow the usage instructions below:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Lightening up to 4 levels	 1:1,5  40	 ***1/3 Lightening Booster 000 + 2/3 colour + 75 g Activator 40 volume	 35 MIN	 20 MIN

*Standard processing time. Processing time can be extended according to hair type, if necessary. **It is possible to use Mask with a heat source: processing time will be reduced by 10-15 minutes. Always check the development of the colour result before colour removal. *** the mixing ratio can be modified to 1/2+1/2 to increase lightening.



Upon completion of the processing time, adding a little water and emulsify the colour.



Rinse thoroughly, then wash with the most appropriate Davines shampoo. Rinse, then apply the most appropriate Davines conditioner.

5.8 INTENSIFIERS

Intensifiers are enhancing and creative new shades, designed to enhance or modify the reflects. They are formulated to be mixed with any Mask with Vibrachrom shades, in accordance with the desired chromatic reflect or on their own for creative uses.

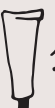
The Intensifiers are the following:

0,66 RED
0,25 VIOLET
0,18 DARK ASH
0,12 LIGHT ASH

When the stylist wants to enhance a reflect, it is suggested to mix the Intensifiers chosen with the most suitable Mask with Vibrachrom shade, following the ratio written above:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Reflect enhancement	 1:1,5 	 1/5 Intensifier (10 g) + 4/5 colour (40 g) + 75 g Activator	 35 MIN	 20 MIN

Whenever the stylist wants to modify a reflect, it is suggested to mix the Intensifiers chosen with the most suitable Mask with Vibrachrom shade, following the ratio written above:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Reflect modification	 1:1,5 	 1 part Intensifier (20 g) + 2 parts colour (40 g) + 90 g Activator	 35 MIN	 20 MIN

Intensifiers are formulated also to be used alone, as a creative tool, directly on previously lightened or bleached hair.

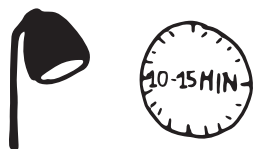
In this case, we suggest following the guidelines inserted in this table:

SERVICE	MIXING RATIO	MIXTURE PER APPLICATION	PROCESSING TIME	
			* 	** 
 Creative use	 1:1,5  10	 50 g Intensifiers + 75 g Activator 10 vol	 35 MIN	 20 MIN

5.9 VIRGIN HAIR APPLICATIONS

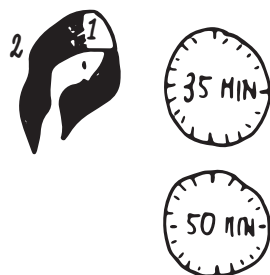
To prepare the mixture, weigh the colour and the correct Activator, following the mixing ratios above.

For lightening, Davines recommends two different application methods, depending on the hair to be treated



All services are based on standard processing time without heat. Mask with Vibrachrom™ can be used with a heat source: processing times are reduced by 10-15 minutes less. Always check the colour result before removing the colour.

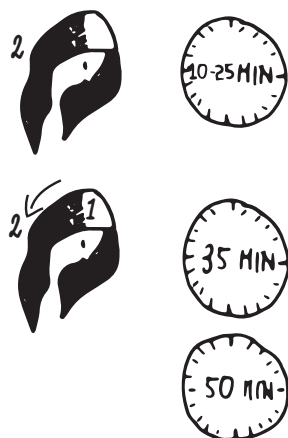
For fine or normal hair.



Apply the mixture throughout the lengths and ends (2), starting 1-2 cm from the root area (1). When completed, immediately prepare another colour mixture but this time with a lower strength Activator and apply to the root area. Process for 35 minutes or 50 minutes for the Extralift series.

For example: if a client with natural level 4 has requested 7,73, apply 7,73 plus Activator 40 volume mixture to the length and ends, then immediately apply 7,73 plus Activator 30 volume to the root area.

For particularly resistant or ethnic hair.



Apply the colour mixture throughout the lengths and ends (2), starting 1-2cm from the root area. Leave it to process for 10-25 minutes.

When this time has passed, then prepare the mixture again, but this time with a lower volume Activator, and apply it from the root area to the ends (1→2). Process for 35 minutes or 50 minutes for the Extralift series.



When darkening virgin hair for the first time, apply the colour mixture directly from the roots and throughout the lengths and ends (1→2) in one application. Leave to process for 35 minutes.

5.10 REGROWTH APPLICATION

To prepare the mixture, weigh the colour and the correct Activator, following the mixing ratios suggested for the desired result.



Apply the colour mixture to the new growth (1). Processing time 35 minutes or 50 minutes for the Extralift series.

5.11 REFRESHING LENGTHS AND ENDS

Minimum color fade, use if colour loss is up to half a level and partial loss of reflect intensity. After standard processing time is complete, at the basin add some water, emulsify the colour through zones 2 & 3 and process for 5 min.

Moderate color fade.

The use of VIEW to refresh Zones 2 & 3 is always our first recommended practice.

Mask with Vibrachrom for moderate colour fade, use if colour loss is up to a level and partial loss of reflect intensity.

- Remember it is not necessary to mix natural base with the reflective nuance when refreshing lengths and ends.
- Mask Shade + 5 vol, 1:2 ratio on damp hair, development time up to 20 min, visually check the colour during processing time
- Recommendation of 5 min up to 10 min for blonde shades again visually check the colour during processing time

Minimum color fade.



Let the colour process on the root for...



Then add some water, emulsify the colour, process for... then rinse

Moderate color fade.



Let the colour process on the root for...



Mix Mask Shade + 5 vol, 1:2 ratio, apply on damp hair, process for...

Moderate color fade, blonde shades



Mix Mask Shade + 5vol, 1:2 ratio, apply on damp hair, process for...

Visually check the colour during processing time.

Extreme colour fade, use if colour loss is more than 2 levels and considerable loss of reflect intensity. The use of VIEW to refresh Zones 2 & 3 is always our first recommended practice. After 15 min development time of Mask, use a VIEW shade with Activator 10 vol, apply to Zones 2 & 3 and process for a further 20 min.

When the hair is extremely faded, the pre-pigmentation service may also be applied to support the reflect of the shade and the durability of the colour result over time. Please refer to the “**Re-pigmentation**” guidelines.

5.12 REPIGMENTATION

We recommend this service when the professional hair colourist needs to darken lightened hair by two or more levels. This method supports a final colour result to be long-lasting and meet chromatic expectations. This technique prepares the hair for the final colour result by replacing missing pigments that were previously removed by lightening or bleaching, or by extensive and repeated exposure to external factors like the sun or swimming pools. Choosing the correct replacement pigment is essential to achieve a great result. The following three Mask with Vibrachrom™ reflective nuances support successful repigmentation:

- 8,33 to support golden colours (yellow)
- 8,44 to support copper colours (orange)
- 6,66 to support red colours (red)



After shampooing, towel dry the hair and apply the repigmentation mixture to all areas that require repigmentation.



Use the mixing ratio: 1 part colour + 2 parts Activators 5 volume. Processing time: 10-20 minutes.

To increase the colour deposit, it is possible to use Activator 10 and 20 volume, instead of Activator 5 volume, depending on the hair conditions. Rinse the colour then dry hair before proceeding to apply the desired colour.

When the desired colour is a natural or a delicate reflect, it is important to remember that repigmentation with 8,33 (yellow) generally supports nuances with levels 6 to 10, while 8,44 (orange) will support nuances with levels 3 to 6, and 6,66 (red) from levels 1 to 3.

8,33 (yellow)		supports from level 6 to 10
8,44 (orange)		supports from level 3 to 6
6,66 (red)		supports from level 1 to 3

Repigmentation can be done with VIEW.

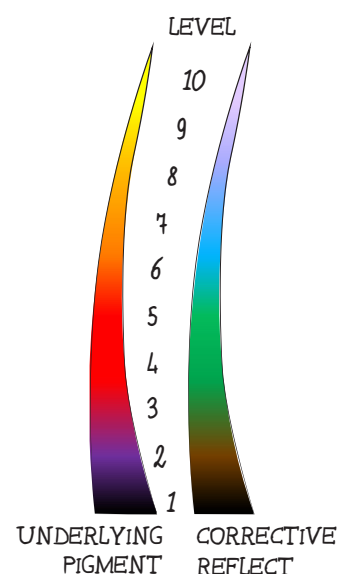
Step 1. Apply with a sponge/brush, on dry clean hair the shade required to support the result you would like to achieve.

Do not use activator and do not rinse. Only apply on the lighter parts of the hair. For warm shades choose a colour/reflect 1 level lighter. For cool shades above level 6 only, choose a colour/reflect 2 levels lighter. It is important not to overload the lighter part of the hair with product.

Step 2. Apply the final colour formula on top of the re-pigmentation, with the activator required and let it process for full time according to Mask processing time.

Natural Results

LEVEL	WARM	COOL
10	-	-
9	-	-
8	9,4	10,21, 10,23
7	8,33	9,32
6	7,4, 7,34	8,23
5	6,44	6,14, 6,74
4	5,34	5,14
3	4,44	4,56
2	-	-
1	-	-



Intense Warm Fashion Results

LEVEL	WARM
10	-
9	-
8	8,44
7	8,44
6	6,66, 6,44
5	5,65
4	5,34
3	4,44
2	-
1	-

TIP! Warm intense fashion results for example, 6,66, 8,44 and 5,56 can also be better supported for a brighter reflect by using a more intense underlying pigment replacement.

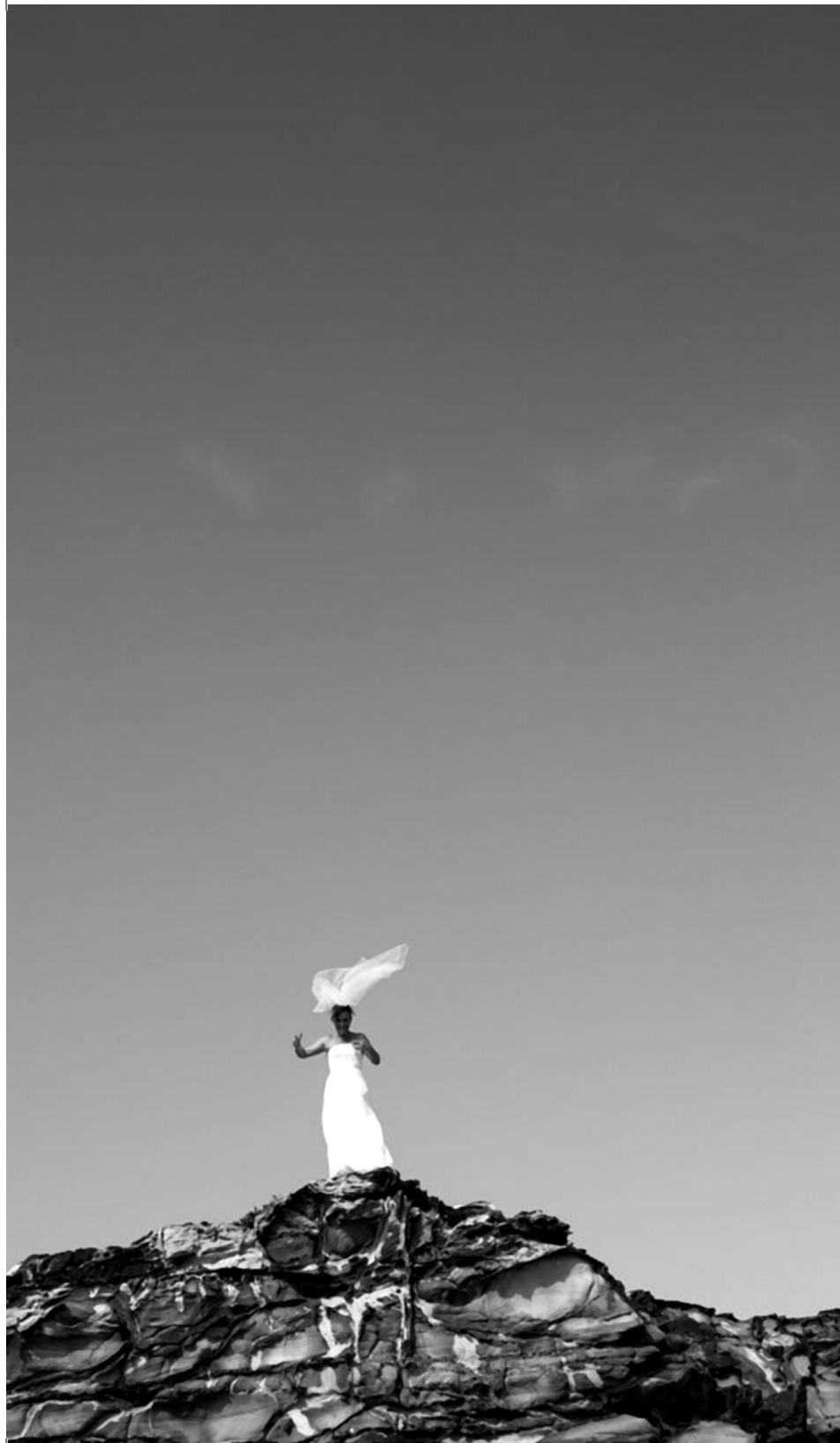
Repigmentation can also be done with Davines Finest Pigments. For porous hair and very lightened hair, when the desired nuance is level 6 and above use either Golden Finest Pigments by itself or a mixture of Golden and Copper. When the desired nuance level is between levels 6 and level 3, use either Finest Pigments Copper by itself or a mixture of Copper and Red. Optionally and if needed, 1/3 of a corresponding Finest Pigments Natural shade can be added to the repigmentation mixture. When the desired nuance is between levels 3 and 1, use either pure Red or a mixture of Red and Copper. Optionally and if needed, 1/3 of a corresponding Finest Pigments Natural shade can be added to the repigmentation mixture.

For porous, lightened hair, use nuances level 6 and above		Finest Pigments Golden or a mixture of Golden and Copper
For hair with a level between 6 and 3		Finest Pigments pure Copper or a mixture of Copper and Red. (option: plus Finest Pigments natural shade)
For hair with a level between 3 and 1		Finest Pigment pure Red or a mixture of Red and Copper (option: plus Finest Pigment natural shade)

5.13 PRE-PIGMENTATION

The Mask with Vibrachrom™ formulation was designed to ensure pre-pigmentation method is no longer necessary.

This technique was generally suggested for unevenly-distributed white hair concentration. Vibrachrom technology, which equalizes pigment distribution, has made it no longer important to use this procedure to achieve perfectly uniform white hair coverage.



6. POST COLOUR CARE

Post-treatment is important to restore the hair's acidic pH after colour services. Davines offers a wide range of products from the MINU and Alchemic lines that can be used as post-colour treatments.

- MINU Essential Haircare offers a complete range of products for coloured hair, without pigments and with active ingredients to keep the hair healthy.
- ALCHEMIC is a specific line to both treat and maintain colour; it consists of a range of shampoos and conditioners enriched with direct pigments for a longer lasting colour between salon visits.

7. TECHNICAL APPENDIX

Colour and colorimetry: basic concepts

7.1 COLOUR

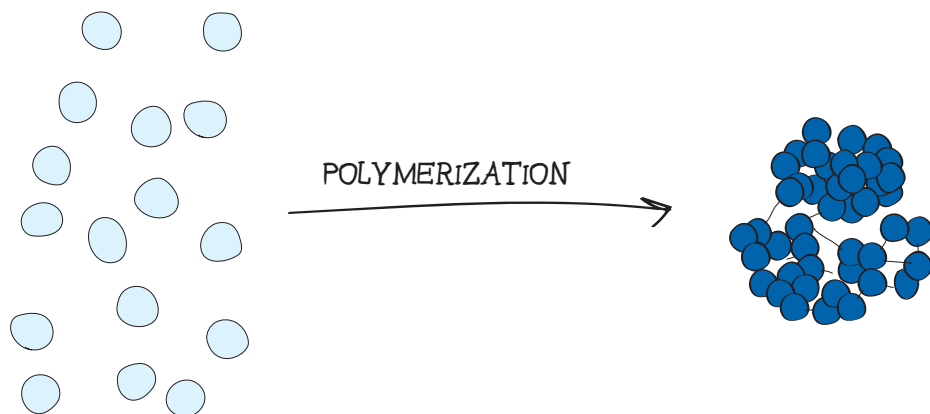
What is colour? The perception of colour is a complex phenomenon and is the result of three processes: physical, chemical and psychological.

Colour is light; in fact, it is created by interactions between light and objects. Light is an electromagnetic wave; its reflection, transmission and, in general, its behaviour are physical processes, governed by laws of physics. Additionally, colour has a chemical component in that the way light interacts with an object depends on the chemical nature of the object itself. Finally, colour is also a psychological process, as our brain elaborates the perception of colour it receives through sight.

Colour, therefore, is a complex phenomenon and, being a subjective perception, it is difficult to describe and define each colour objectively. The science of colorimetry explains the theory of colour and there are several international classification systems which, using physical-chemical parameters, provide an unequivocal, universally accepted definition for every colour that exists in nature. Below are a few basic concepts of colorimetry. Generally speaking, these concepts can also refer to hair colouring theory, though not always. In fact, hair colouring is influenced by other factors, which vary from subject to subject like hair structure and melanin composition, which we will discuss later, and pigments, which are not coloured molecules but become so only during colouring. Therefore, with regards to hair colouring, assessments must be less scientific and more subjective.

7.2 BASIC PRINCIPLES OF COLOURING

The pigments used in oxidation colours are complex substances, chemically different and colourless. To create a colour, they need to bind and form a bigger molecule in a process known as polymerization, which is triggered by the oxygen released by the hydrogen peroxide used during the colouring process. This is why it is also called oxidation reaction.



Hydrogen peroxide is an acid substance made up of two atoms of hydrogen and two atoms of oxygen. Its molecule (H_2O_2) is highly unstable and spontaneously releases oxygen thus turning into a water molecule (H_2O). Free oxygen is very active and tends to easily bond with other substances, giving rise to reactions known as oxidations.

During the colour service, the oxygen released by the hydrogen peroxide contained in the activator carries out three functions:

- it **OXIDISES** the pigments to produce the final colouring substances (polymerization reaction).
- it **LIGHTENS** natural melanin. The lightening power is closely related to the volume (or concentration) of the hydrogen peroxide used.
- it **BALANCES** the pigment concentration and the optimal pH for the colouring mix.

Thanks to all these actions, hydrogen peroxide is the starting point of the whole hair colouring process.

The concentration (or titration) of hydrogen peroxide is usually expressed in "volumes": e.g. 20 volumes correspond to 6% of H_2O_2 .

The right volume is chosen according to the desired result:

ACTIVATOR	% H_2O_2	EXPECTED RESULT WITH COLOUR
5 volume	1,5	Toning bleached or coloured hair, hair with highlights or refreshing natural hair, length and ends
10 volume	3	Darkening
20 volume	6	Covering white hair
20 volume	6	Changing reflect or lightening by a level
30 volume	9	Lightening by two levels
40 volume	12	Lightening up to three levels and 5 levels with extralift

Along with pigments and hydrogen peroxide, oxidation colouring requires an alkalizing agent to open the hair cuticles and enable pigment penetration.

The alkalizing agent normally used in "traditional" oxidation colourings is ammonia, which is a gas found in nature (NH_3), therefore volatile and distinguished by an unmistakable odour. In cosmetic formulations it is used in an aqueous solution (NH_4OH), called ammonium hydroxide. Ammonia is a substance that is found in nature, even inside the human body, where it carries out important biological functions.

Ammonia has two functions in the colouring process:

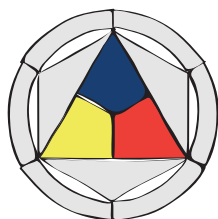
1. It speeds up reactions, favouring oxygen release from the hydrogen peroxide, thereby catalysing the polymerisation reaction.
2. It creates an alkaline environment, which opens the cuticle keratin scales, allowing the pigments to penetrate.

Being a gas, ammonia evaporates quickly, unlike other alkaline substances. This means the physiological pH is quickly restored once the colour service is completed.

7.3 PRIMARY, SECONDARY AND TERTIARY COLOURS

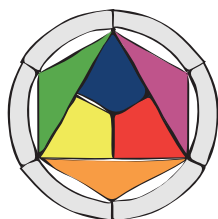
According to colorimetry, colours can be divided into primary, secondary and tertiary colours. Primary colours get their name from the fact that they are not created by mixing other colours. There are three:

- blue
- yellow
- red



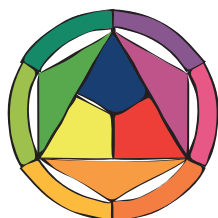
An infinite number of other colours can be made by mixing the primaries together. Mixing primary colours in equal parts yields secondary colours, which are:

- red + blue = violet
- red + yellow = orange
- yellow + blue = green



Mixing primary colours with secondary colours yields tertiary colours:

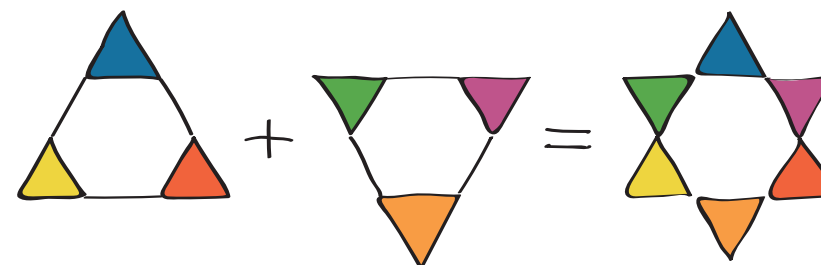
- yellow + orange = yellow-orange
- red + orange = red-orange
- red + violet = red-violet
- blue + violet = blue-violet
- blue + green = blue-green
- yellow + green = yellow-green



When we talk about primary colours in hair colouring, we are essentially referring to red, blue and yellow and not magenta, which is a purplish red. This is because we refer to the type of melanin in the hair which, as we will see, is a chemically complex biological structure that makes it difficult to accurately identify the tone.

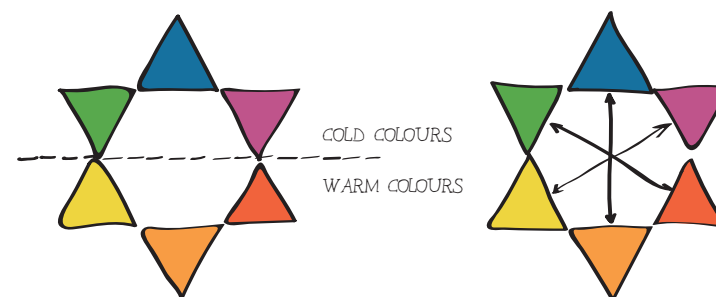
7.4 OSTWALD'S COLOUR STAR

The Ostwald star is a useful tool to explain and understand the properties of colours. It can be created by drawing two intersecting triangles: the primary colours go in the corners of the first triangle, while the secondary colours, obtained by mixing the primary colours, go in the corners of the second triangle.



This star gives us much more information on colours.

In fact, we can distinguish the primary (yellow, red and blue) and secondary colours, as done previously, highlight warm and cool colours, and easily understand the concept of "complementarity". Dividing the star horizontally, at the top we have green, blue and violet, and at the bottom, yellow, orange and red. The former (violet, green and blue) are considered cool colours, while red, orange and yellow are considered warm.



The concept of complementarity is one colour's capacity to contrast or neutralise another colour. To be precise, the colours across from each other in the Ostwald star - blue and orange, purple and yellow, red and green - are complementary colours, that is, one tends to neutralise the reflect of the other.

This property is very useful and is used in colouring hair to reduce an excessively intense or undesired reflect. For example, to contrast an excessively intense red reflect, we can use a nuance containing green pigments. Or, to contrast yellow when lightening, we can use violet, always keeping in mind that cool tones can contrast warm ones but not vice versa, as the final result would be an overlapping of colours with a metallic effect.

7.5 CLASSIFICATION SYSTEM

The Mask with Vibrachrom™ classification system borrows from international numbering, which defines both colour levels and tone using numbers. Every nuance code contains three numbers. For example, in the nuance 6,73, the first number, in this case the 6, corresponds to the base level, that is, how dark or light the hair colour is. The first number always refers to how light or dark a nuance is. The colour levels can be:

1 black	6 dark blonde
2 dark brown or very dark brown	7 blonde
3 medium brown	8 light blonde
4 light brown	9 very light blonde
5 lightest brown	10 lightest blonde - platinum

The levels represent “neutral” tones, that is, they represent brown without adding red, yellow, blue or other components. The second number, in this case the 7, corresponds to the primary or main reflect, needed to define the key chromatic character of a nuance, also indicating whether it is a cool or warm nuance. The third number, in this case the 3, corresponds to the secondary reflect. The possible reflects indicated by the second and third numbers are:

1 blue / ash
2 Irisé (purple)
3 gold
4 copper
5 mahogany
6 red
7 beige (between yellow and green)
8 green

Among the international numbers used to indicate reflects, we also have number 7, which in the Mask with Vibrachrom™ classification system identifies the beige reflect, and the new number 8, which indicates that has a green-based chromatic characteristic.

Now that we know what possible reflects there are, we can identify the colour combinations that create them. It is easy and intuitive for some of them:

- golden reflects have a yellow base
- reds have a red base
- copper reflects have an orange base (yellow and red)
- 8 reflects have a green base

Other reflects are less intuitive, like ash reflects, which have a blue base, 2 reflects have cool violet reflect, based upon violet and blue; mahogany reflects which are, 5 have a combination of violet and red; and beige reflects are structured by combining green and yellow. The tertiary colours in Mask with Vibrachrom™, then, are:

- violet + blue = irisé
- violet + red = mahogany
- green + yellow = beige

NOTES

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

NOTES

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